CardioTwin: Real-Time Biophysical & Neural Digital Twin Architecture

End-to-End Computational Pipeline: Multi-Scale Telemetry → Biophysical Solvers → AI Surrogate → Clinical Decision Support

1. INGESTION LAYER

- Continuous Lead-II ECG (250 Hz)
- Non-invasive Blood Pressure (SBP/DBP)
- Pulse Oximetry (SpO2) & Heart Rate
- EHR Demographics & BSA (m^2)
- Baseline NYHA Functional Class
- Cardiac Biomarkers (Troponin, BNP)

2. BIOPHYSICAL ODE CORE

- Suga-Sagawa Elastance E(t)
- Double-Hill Activation e_n(t/T)
- 3-Element Windkessel Arterial ODE
- Forward Diode Valve Fluid Dynamics
- Ventricular Volume Conservation
- Real-Time Runge-Kutta Numerical Solver

3. AI / ML SURROGATE LAYER

- MultiOutput Neural Surrogate Model
- Instantaneous Inference (<1.5 ms)
- Gradient Boosted Arrhythmia Classifier
- 6-Class Conduction Detection (AFib, VT)
- Composite Decompensation Risk (HDRI)
- Gini / Tree SHAP Feature Importance

4. CLINICIAN INTERACTION

- Glassmorphic Streamlit UI
- 3D Pulsatile Ventricular Mesh (Plotly)
- Dynamic Multi-Wave Wiggers Diagram
- Dual Baseline-vs-Intervention PV Loops
- In-Silico 'What-If' Drug Titration
- Automated EHR Clinical Dossier Export

CLINICAL SAFETY, ETHICAL GOVERNANCE & REGULATORY PROTOCOLS

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